

Eastern Equine Encephalitis, an Overview

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If you have grown up pretty much anywhere, you have been exposed to, bitten by, and annoyed by mosquitoes. But this can come at a cost, which has been recently knocking on our very doors. In later months, scientists have discovered a resurgence of *Eastern equine encephalitis*.¹ This has led to much concern and questions about what exactly the disease is, and



in this article, we hope to eliminate some of these questions.

Triple E is a mosquito born illness,² meaning it is spread through the bite of an infected mosquito. This disease as of right now, has no known treatments or medications, which is partially why it is considered such a serious ailment. It is found almost exclusively on the east coast of the United States, (hence the name) and is most prevalent in Massachusetts.³ This disease has a 30% mortality rate, meaning that if contracted, there is a 30% chance that the victim will die,⁴ And an



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<https://www.bedfordma.gov/CivicAlerts.aspx?AID=371#:~:text=August%2016%2C%202024%2D%20The%20Massachusetts%20identified%20in%20Worcester%20County.>

² <https://www.cdc.gov/eastern-equine-encephalitis/about/index.html>

³ <https://www.cdc.gov/eastern-equine-encephalitis/data-maps/current-year-data.html>

⁴ <https://www.cdc.gov/eastern-equine-encephalitis/about/index.html>

incubation period of around 4-10 days.⁵ What this means is that if the disease is contracted, there is a period of 4-10 days in which an infection has not yet developed, but the disease is still within the body.⁶

Triple E can take place in two forms: *febrile* and *neurological* disease.⁷ Febrile disease takes place with symptoms that greatly resemble that of the flu. This includes fever, joint pain, and body aches. Typically, it lasts 1-2 weeks and a full recovery can be made if it does not develop into *neurological disease*.⁸ *Neurological Disease* is the most infamous form of Triple E. This disease can take place in two forms: *meningitis* (when the membranes around the spinal cord and brain becomes inflamed) and *encephalitis* (when the brain itself becomes inflamed).⁹ This is the most fatale type of Triple E and has a relatively high mortality rate. Symptoms include vomiting, diarrhea, fever, headache, coma, seizures, behavioral changes and drowsiness. In infants, Neurological Disease takes place soon after the initial contraction. In adults, Encephalitis may take a few days of systemic illness before it takes place.¹⁰

But what do all these morbid numbers and facts have anything to do with BISB? And why is the disease so inconspicuous, yet so extremely dangerous. We spoke with biology teacher, Mrs. Dolan who stated, “If the disease was more well known, then it would be much easier to prevent.” This is true, as many people are unaware of what Triple E is and are frankly quite uninterested. But if people do not become aware of it, they will be unable to implement strategies to prevent it in their daily lives. Some strategies people can implement include not staying out late (mosquitoes are most active during dusk), wearing insect repellent, avoid spending prolonged periods of time near stagnant water, and covering areas of the body exposed to insects.

So, what. What does this mean? Don’t ever go outside, avoid woods, and always wear long sleeves? No, not exactly. Whilst it is important to be cautious and preventative, it is also important to allow oneself to live comfortably and remember that Triple E is rare disease that is preventable and should not cause much anxiety and fear in the public.

⁵ <https://www.cdc.gov/eastern-equine-encephalitis/about/index.html>Was

⁶ <https://my.clevelandclinic.org/health/articles/incubation-period>

⁷ <https://www.cdc.gov/eastern-equine-encephalitis/about/index.html>

⁸ <https://www.cdc.gov/eastern-equine-encephalitis/about/index.html>

⁹ <https://www.cdc.gov/eastern-equine-encephalitis/about/index.html>

¹⁰ <https://www.cdc.gov/eastern-equine-encephalitis/about/index.html>

Sources:

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